

§6 Improper Integrals

* The definite integral $\int_a^b f(x) dx$ where the interval is infinite or f has an infinite discontinuity in $[a, b]$ is called an **improper integral**.

Type 1: Infinite Intervals

Definition

(a) $\int_a^t f(x) dx$ exists for every number $t \geq a$

$$\Rightarrow \int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

(b) $\int_t^b f(x) dx$ exists for every number $t \leq b$

$$\Rightarrow \int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$$

• The improper integrals $\int_a^\infty f(x) dx$ and $\int_{-\infty}^b f(x) dx$ are called convergent if the corresponding limit exists and divergent if the limit does not exist.

(c) $\int_a^\infty f(x) dx$ and $\int_{-\infty}^a f(x) dx$ are convergent

$$\Rightarrow \int_{-\infty}^\infty f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^\infty f(x) dx$$

where a can be used for any real number.

• If the improper integral $\int_a^\infty f(x) dx$ is convergent, then it is interpreted as the area under the graph of $y = f(x)$ for $x \geq a$.

Exam 1 Determine whether the integral $\int_1^{\infty} \frac{1}{x} dx$ is convergent or

divergent.

(sol)

(cf) $\int_1^{\infty} \frac{1}{x^2} dx =$

Exam 2 Evaluate $\int_{-\infty}^0 x e^x dx$.

(sol)

Exam 3 Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$.

(sol)

Exam 4 For what values of p is the integral

$$\int_1^{\infty} \frac{1}{x^p} dx$$

convergent?

(sol)

(Exam) Determine whether the integral $\int_0^{\infty} \cos x dx$ is convergent or divergent.

(sol)

Type 2: Discontinuous Integrands

Definition

(a) f is continuous on $[a, b)$ and is discontinuous at b .

$$\Rightarrow \int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

(b) f is continuous on $(a, b]$ and is discontinuous at a .

$$\Rightarrow \int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$$

• The improper integrals $\int_a^b f(x) dx$ is called convergent if the corresponding limit exists and divergent if the limit does not exist.

(c) f has a discontinuity at c , where $a < c < b$ and both $\int_a^c f(x) dx$

and $\int_c^b f(x) dx$ are convergent

$$\Rightarrow \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

Exam 5 Find $\int_2^5 \frac{1}{\sqrt{x-2}} dx$.

(sol)

Exam 6 Determine whether $\int_0^{\frac{\pi}{2}} \sec x \, dx$ converges or diverges.

(sol)

Exam 7 Evaluate $\int_0^3 \frac{dx}{x-1}$ if possible.

(sol)

*** Comparison Theorem**

f, g : continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.

(a) $\int_a^\infty f(x) dx$ is convergent $\Rightarrow \int_a^\infty g(x) dx$ is convergent.

(b) $\int_a^\infty g(x) dx$ is divergent $\Rightarrow \int_a^\infty f(x) dx$ is divergent.

Exam 8 Show that $\int_0^\infty e^{-x^2} dx$ is convergent.

(sol)

Exam 9 The integral $\int_1^{\infty} \frac{1+e^{-x}}{x} dx$ is divergent by the Comparison

 Theorem because $\frac{1+e^{-x}}{x} > \frac{1}{x}$ for all $x \geq 1$ and $\int_1^{\infty} \frac{1}{x} dx$ is
 divergent.

(Exam)

1. $\int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx =$

2. $\int_0^{\sqrt{2}a} \frac{2x}{(x^2-a^2)^{2/3}} dx \quad (a > 0)$

$$3. \quad \int_{-\infty}^{\infty} \frac{1}{x^2} dx$$

$$4. \quad \int_0^1 \ln x \, dx =$$